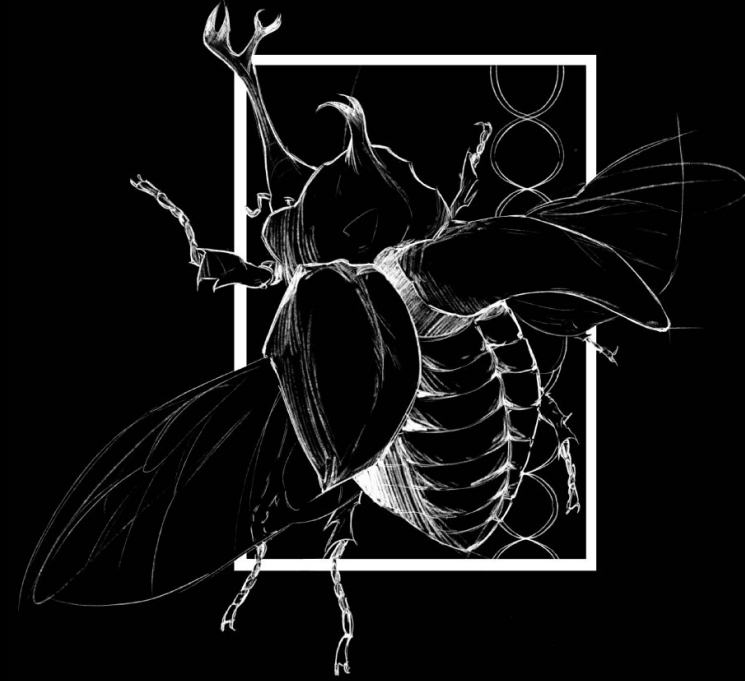




Background

Gymnosperms are vascular plants that have an exposed seed or ovule. Within gymnosperms, there are four divisions: Pinophyta, Cycadophyta, Ginkgophyta, and Gnetophyta. The big question of this research is this: is genome size a predictor for wood density in gymnosperms? This is an interesting comparison to check because the genome size can impact how big a plant cell can be. This traces to an important concept where cells have to maintain an optimal surface area-volume ratio to compartmentalize their necessary functions. If we scale this up, the bigger the genome size, the bigger the cell will. This also means that bigger cell size means the less of the number of those cells will be in that plant.

Because genome size can impact the number of cells, it also affects the wood density. Wood density refers to how much wood is on the plant. My hypothesis for this question is that the bigger the genome size, the lesser the wood density. The reason why is that because a genome size will give a larger cell. A larger cell means there will be less to fit in a confined space. The less space there is means that there is less weight. This in turns gives a lesser wood density. With the current data, wood density data is between 0.21-0.65 g/cm³ while the genome size data is between 3.87-36 1c pg. This is a broad set of values that can support the big question of this research.

A comparative approach on studying this relationship is necessary because gymnosperms has a vast evolutionary history. Gymnosperms has over 300-350 million years of evolutionary history and spans over 30,000 thousand species. As such, comparative methods can account for the shared ancestry. Additionally, it can distinguish between correlated evolutions over coincidental similarity.

For this study, a Phylogenetic Generalized Least Squares (PGLS) analysis was done to test the evolutionary correlation between genome size and wood density. Additionally, a model comparison was done between Brownian Motion (BM) and Ornstein-Uhlenbeck (OU) models to see whether the traits were correlated or was based on random drift.

Study Design & AI

The use of AI was integral for this study. All AI use was done through chat.tamuai. AI was initially used for the brainstorming of the ideas behind this study. Additionally, AI was used to write R scripts needed for this study. Initially, the use of AI was seen in the R script to make a database from the species list, the wood density, and the genome size list. It was pruned down to all species that had wood density and genome size available. Additionally, AI creates R scripts to make the PGLS plot and the BM vs OU model comparison.



Cycas revoluta - softest wood with wood density of 0.21 g/cm³



Halocarpus biformis - hardest wood with wood density of 0.65 g/cm³

Results

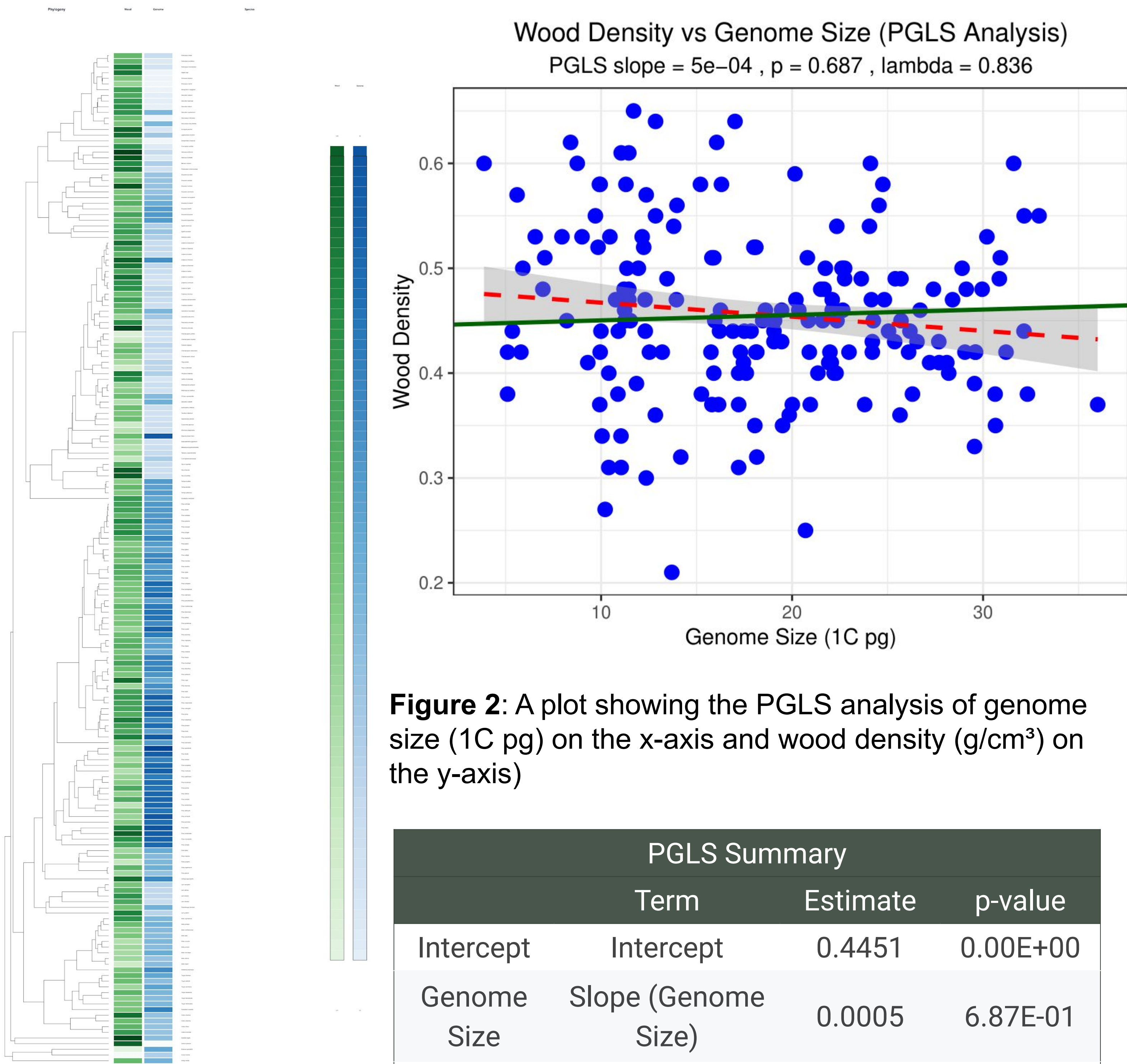


Figure 1: A pruned evolutionary tree of gymnosperms with shaded boxes corresponding to the traits of genome size and wood density.

Figure 2: A plot showing the PGLS analysis of genome size (1C pg) on the x-axis and wood density (g/cm³) on the y-axis)

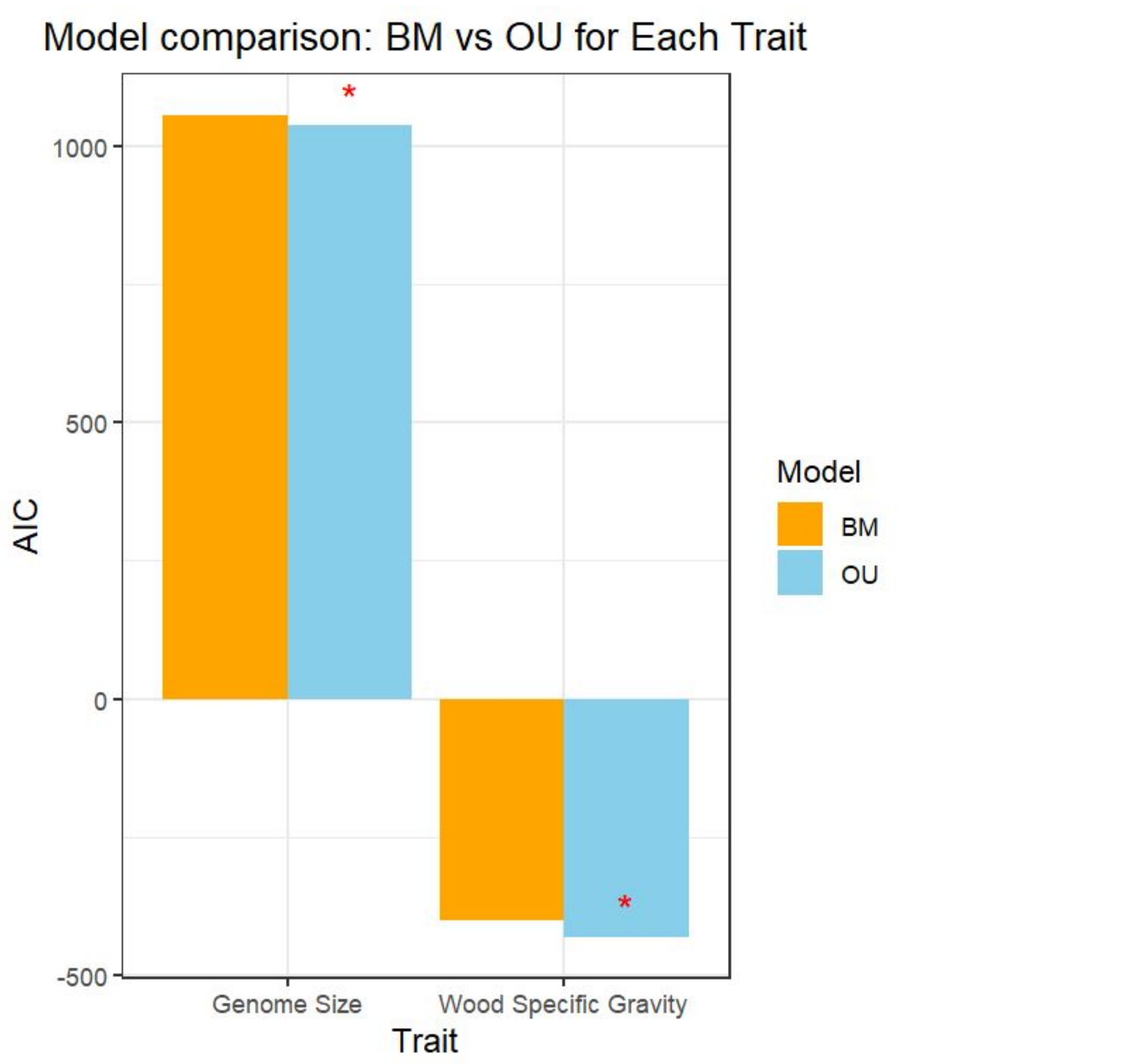


Figure 3: A bar graph comparison showing the results between BM and OU models for both traits of genome size and wood specific gravity (wood density)

Discussion

The data collected doesn't support the original hypothesis. However, the data doesn't fully disapprove the data. The PGLS data summary is a good explainer of this conclusion. The slope (the green line in the PGLS plot) is a slightly small positive number. However, the R-squared value is also a small value, indicating that there is no relationship between the genome size and wood density. Additionally, the p-value further proves that there isn't a significant relationship between these two traits. Despite that, the high lambda value indicates that both traits are strongly influenced by shared common ancestry. Based on the phylogenetic tree, there is variation in the relationship between genome size and wood density. There are some species that support the original hypothesis while some disapprove. In the end though, due to this mix of species correlation between the two traits has no relationship overall. Within each clades of the phylogeny, the traits are somewhat conserved. That means due to shared common ancestry, species in these clades often inherit similar traits. However, there is high within-group variation. While ancestry influences, there are other factors involved such as environmental influences, mutations, and adaptations that cause this variation. The results that were discussed can be explained by the model comparison. Regarding genome size, it had a higher value for Brownian Motion (BM). This means that the variations in genome size are influenced by random drift, meaning this trait is less constrained. However, wood density scored high on the Ornstein-Uhlenbeck (OU) model. This means that wood density was more evolutionary constrained. This is strong evidence that the trait values are pulled towards more stabilizing influence.

Conclusions & Acknowledgements

There is no evolutionary relationship between wood density and genome size in gymnosperms. The reason for this is that wood density is evolutionary constrained while genome size is more influenced by random drift. Despite this lack of correlation, this correlation is better found within specific clades of gymnosperms, where each traits are somewhat conserved to be nearly identical to the common ancestor. Acknowledgements: I would like to thank Dr. Blackmon for the opportunity to do this poster and giving me the opportunity to do research here at Texas A&M University. I would also like to acknowledge chat.tamuai in creating this statistical analysis done for this poster using the models proved by ChatGPT and Claude.